

Metric Topology of $\mathbb{C}$

Complex Analysis

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Definition

A complex number z may be thought of as coordinates in the $\mathbb{R} \times i\mathbb{R}$ plane. Such plot is called an **Argand Diagram** and is a common representation of complex numbers.

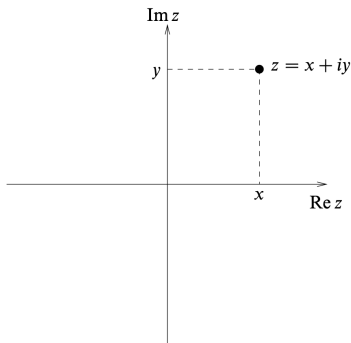


Figure 1: The Argand diagram of a point $z \in \mathbb{C}$

Remark

The addition of complex numbers can also be thought as the sum of coordinates in the $\mathbb{R} \times i\mathbb{R}$ plane.

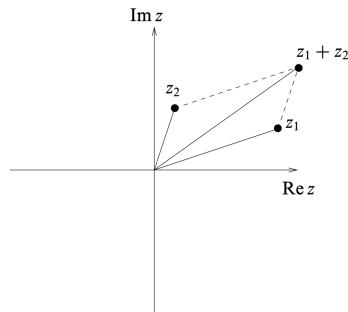


Figure 2: The addition of two complex numbers

Remark

The complex numbers $\pm 1, \pm i$ have modulus unity and using the fact that $|z_1 z_2| = |z_1| |z_2|$ we can see immediately that multiplying a complex number z by one the former, then the latter has the same modulus as z (we identify this operation as a rotation, fact which we will justify later).

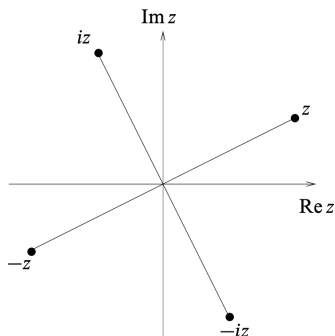


Figure 3: Multiplication of z by $\pm 1, \pm i$

Remark

Complex conjugation corresponds to a reflection of z in the real axis of the Argand diagram.

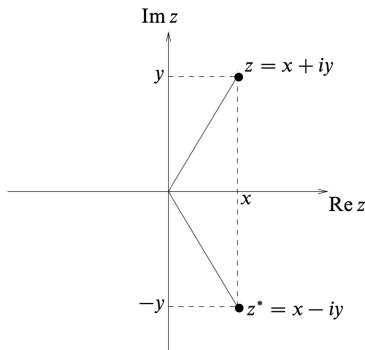


Figure 4: The complex conjugate as a mirror image in the real axis

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Definition

A **sequence** of complex numbers (or a **sequence in $\mathbb{C}$**) $Z := (z_n)$ is a function $Z : \mathbb{N} \rightarrow \mathbb{C}$ with domain $\mathbb{N}$ and range in $\mathbb{C}$

Remark

If $Z : \mathbb{N} \rightarrow \mathbb{C}$ is a sequence, we will usually denote the value of Z at n by the symbol z_n rather than using the function notation $Z(n)$. The values z_n are also called the **terms** or the **elements** of the sequence. We will denote this sequence by the notations

$$Z, \quad (z_n), \quad (z_n : n \in \mathbb{N}).$$

Definition

Let $\varepsilon > 0$, then the **disk of radius ε centred at z_0** is the set

$$D_\varepsilon(z_0) := \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}.$$

A **neighborhood** of z_0 is any set $V \subseteq \mathbb{C}$ that contains a disk of radius $\varepsilon > 0$ centred at z_0 for some $\varepsilon > 0$. We may also refer to $D_\varepsilon(z_0)$ as the **ε -disk of z_0** .

Definition

A sequence $Z = (z_n)$ in $\mathbb{C}$ is said to **converge** to $z \in \mathbb{C}$, or z is said to be a **limit** of (z_n) , if for every $\varepsilon > 0$ there exists a natural number K such that $x_n \in D_\varepsilon(x)$ for all $n \geq K$.

If a sequence has a limit, we say that the sequence is **convergent**; if it has no limit, we say that the sequence is **divergent**.

Theorem

Let $Z = (z_n)$ be a sequence of complex numbers, and let $z \in \mathbb{C}$. The following statements are equivalent.

- a) Z converges to z .
- b) For every ε , there exists a natural number K such that for all $n \geq K$, the terms z_n satisfy $|z_n - z| < \varepsilon$.
- c) For every ε -disk $D_\varepsilon(z)$ of z , there exists a natural number K such that for all $n \geq K$, the terms z_n belong to $D_\varepsilon(z)$.

Remark

When a sequence has a limit z , we will use the notation

$$\lim Z = z \quad \text{or} \quad \lim(z_n) = z.$$

We will sometimes use the symbolism $z_n \rightarrow z$, which indicates the intuitive that the values z_n “approach” the number z as $n \rightarrow \infty$.

Definition

A sequence $Z = (z_n)$ of complex numbers is said to be **bounded** if there exists a real number $M > 0$ such that $|z_n| < M$ for all $n \in \mathbb{N}$.

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Theorem

A convergent sequence in $\mathbb{C}$ is bounded.

Theorem

Let $Z = (z_n)$ and $Z' = (z'_n)$ be sequences in $\mathbb{C}$ that converge to z, z' , respectively, and let $\lambda \in \mathbb{C}$.

- a) Then the sequences $Z + Z' := (z_n + z'_n)$, $Z - Z' := (z_n - z'_n)$, $ZZ' := (z_n z'_n)$, and $\lambda Z := (\lambda z_n)$ converge to $z + z'$, $z - z'$, zz' , and λz , respectively.
- b) If $W = (w_n)$ is a sequence of nonzero complex numbers that converges to w and if $w \neq 0$, then the quotient sequence $Z/W := (z_n/w_n)$ converges to z/w .

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Theorem

Let the sequence $Z = (z_n)$ converge to z . Then the sequence $(|z_n|)$ of moduli converges to $|z|$. That is, if $z = \lim(z_n)$, then $\lim(|z_n|) = |z|$.

Theorem

Let $Z = (z_n)$ be a sequence in $\mathbb{C}$ and let $z \in \mathbb{C}$. If $X = (x_n)$ is a sequence of positive real numbers such that $\lim(x_n) = 0$, and if for some constant $C > 0$ and some $m \in \mathbb{N}$ we have

$$|z_n - z| \leq Cx_n \quad \text{for all } n \geq m,$$

then it follows that $\lim(z_n) = z$.

Definition

Let $z_0 \in \mathbb{C}$ and $\varepsilon > 0$. Then the **open ball** $B_\varepsilon(z_0)$ of radius ε , centred at z_0 is the set

$$B_\varepsilon(z_0) := \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}.$$

Sometimes we may refer to $B_\varepsilon(z_0)$ as the ε -ball of z_0 .

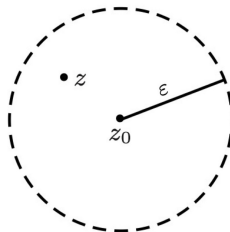


Figure 1: The ε -ball of z_0

Definition

Let $\Omega \subseteq \mathbb{C}$. Then we say that a point $z_0 \in \Omega$ is an **interior point** of Ω if there exists a ball of radius $\rho > 0$ centred at z_0 contained in Ω , that is, $B_\rho(z_0) \subset \Omega$. The **interior** of Ω , denoted by Ω° is the set of all interior points of Ω .

Definition

Let $\Omega \subseteq \mathbb{C}$. We say that Ω is **open** if for each point $z_0 \in \Omega$, there exists a ρ -ball $B_\rho(z_0)$ of z_0 such that $B_\rho(z_0) \subset \Omega$, that is, if each point of Ω is an interior point of Ω .

Definition

Let z_1, z_2 be two points in the complex plane. The **line segment ℓ joining z_1 and z_2** is the set

$$\ell := \{z \in \mathbb{C} : z = z_1 + t(z_2 - z_1), 0 \leq t \leq 1\}.$$

Now let $\{z_i\}_{i=1}^{n+1}$ be $n+1$ points in $\mathbb{C}$. For each $k = 1, 2, \dots, n$ let ℓ_k denote the line segment joining z_k and z_{k+1} . Then the **polygonal path joining z_1 and z_{n+1}** is the set $\bigcup_{i=1}^n \ell_i$. Further, if Ω is a subset of $\mathbb{C}$ and if $\bigcup_{i=1}^n \ell_i \subseteq \Omega$, then we say that the polygonal path joining z_1 and z_{n+1} **lies entirely in Ω** .

Definition

A subset Ω of $\mathbb{C}$ is said to be **connected** if every pair of points z_1, z_2 in Ω can be joined by a polygonal path that lies entirely in Ω . A set that is both open and connected is said to be a **domain** (or a **region**).

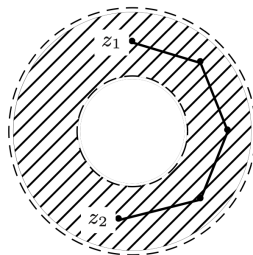


Figure 2: A connected set

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Definition

A set $\Omega \subseteq \mathbb{C}$ is said to be **closed** if its complement, denoted by $\Omega^{\sim}$, is open.

Definition

Let $\Omega \subseteq \mathbb{C}$. A point $\omega \in \Omega$ is a **cluster point** (or a **limit point**) of Ω if for every $\rho > 0$ there exists at least one point $z \in \Omega$ such that $0 < |z - \omega| < \rho$. Equivalently, we may say that ω is a cluster point of the set Ω if every ρ -ball $B_\rho(\omega)$ of ω contains at least one point of Ω distinct from ω .

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Theorem

A complex number ω is a cluster point of a subset Ω of $\mathbb{C}$ if and only if there exists a sequence (z_n) in Ω such that $\lim(z_n) = \omega$ and $z_n \neq \omega$ for all $n \in \mathbb{N}$.

Definition

Let $\Omega \subseteq \mathbb{C}$. A point z_0 is said to be an **exterior point** (or an **isolated point**) of Ω if there is some $\rho > 0$ for which the ρ -ball of z_0 not including z_0 is contained in $\Omega^\sim$ or, that is, $B_\rho(z_0) \setminus \{z_0\} \subset \Omega^\sim$ (the ρ -ball of z_0 does not contain any points of Ω distinct from z_0).

Definition

If $\Omega \subseteq \mathbb{C}$ and $z_0 \in \mathbb{C}$, we say that z_0 is a **boundary point of Ω** if every ball centred at z_0 contains at least one point of Ω and one point in $\Omega^\sim$.

Thus, a boundary point is neither an interior point nor an exterior point.

The set of all boundary points of Ω is denoted by $\partial\Omega$ and is called the **boundary (or frontier) of Ω** .

Definition

The **closure** of $\Omega \subset \mathbb{C}$ is the set

$$\overline{\Omega} := \Omega \cup \partial\Omega.$$

Definition

A set $\Omega \subseteq \mathbb{C}$ is said to be **bounded** if there exists $M > 0$ such that $|z| < M$ whenever $z \in \Omega$, that is, if there exists a ball B of radius M centred at the origin and containing Ω . If Ω is bounded, we define the **diameter** of Ω to be the number

$$\text{diam}(\Omega) = \sup_{z, w \in \Omega} |z - w|.$$